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ORIGINAL ARTICLE

"PLS-SEM: indeed a silver bullet" – retrospective observations and recent advances

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ABSTRACT
In 2011, the *Journal of Marketing Theory & Practice* published "PLS-SEM: Indeed a silver bullet," which became a cornerstone contribution in marketing. Critical reflection of research work is a fundamental building block of science, including one's own writing. In this spirit, we offer a review of our own 2011 paper, assuming we were reviewers with today's background knowledge of the method. Taking a reviewer's perspective in our comments, we clarify several ambiguities in our initial overview presentation and offer updates of – from today's perspective– outdated descriptions that led to some criticisms in recent years which have since been overcome.

Foreword

In 2011, the *Journal of Marketing Theory and Practice* published "PLS-SEM: Indeed a silver bullet" (Hair et al., 2011), which provided a nontechnical primer on partial least squares structural equation modeling (PLS-SEM). The article discusses the method's main characteristics, the underlying algorithm, and the features that distinguish PLS-SEM from covariance-based SEM (CB-SEM; see also, Jöreskog & Wold, 1982), which many researchers have considered the standard approach for estimating models with latent variables. Hair et al.'s (2011) article has had a considerable impact on the dissemination of the PLS-SEM method as evidenced by its massive citation count (the most highly cited article to appear in *Journal of Marketing Theory & Practice*). As of January 2022, the article has been cited more than 15,000 times in Google Scholar and more than 6,000 times in the Web of Science, primarily in business and economics. The article's reach extends to many other research areas, however, such as computer sciences, engineering, environmental sciences, medicine, political sciences, psychology, and sociology. Figure 1 shows a tree map of all research areas with 100 or more citations in the Web of Science, and the corresponding citations as of January 21, 2022.

Hair et al.'s (2011) article has also triggered sometimes fierce debates with critics claiming that it is an advocacy paper (Rönkkö et al., in press) and aggressively promoting (Rönkkö & Evermann, 2013) a supposedly known to be biased estimator (Evermann & Rönkkö, in press). Numerous other scholars have countered the critics on the grounds of analytical evidence (e.g. Cook & Forzani, 2020; Rigdon, 2012), simulation findings (e.g. Hair et al., 2017a; Henseler et al., 2014; Sharma et al., in press-a), and conceptual arguments (e.g. Rigdon et al., 2017; Sarstedt et al., 2016). Russo and Stol (in press) observe an increasingly polarized debate regarding PLS-SEM's efficacy, in which critics oftentimes disregard recent developments in the field. Petter (2018, p. 10) summarizes such critiques as "haters gonna hate," concluding that scholars should "feel the love for PLS."

In light of these developments, revisiting the article that has triggered some of these debates is timely and warranted. We do so by taking a reviewer's perspective that evaluates Hair et al. (2011) by today's standards. The objective of this article is to clarify several ambiguities in our initial overview presentation and offer updates of – from today's perspective – outdated descriptions that led to some criticism in recent years which has since been overcome. In doing so, we contribute to advancing the knowledge of the PLS-SEM method and ensuring researchers can adequately exploit its valuable properties in their research.

Introduction: Dear Authors ...

Thank you very much for giving us the opportunity to review your paper. PLS-SEM has gained considerable attention in recent years. In fact, PLS-SEM use is

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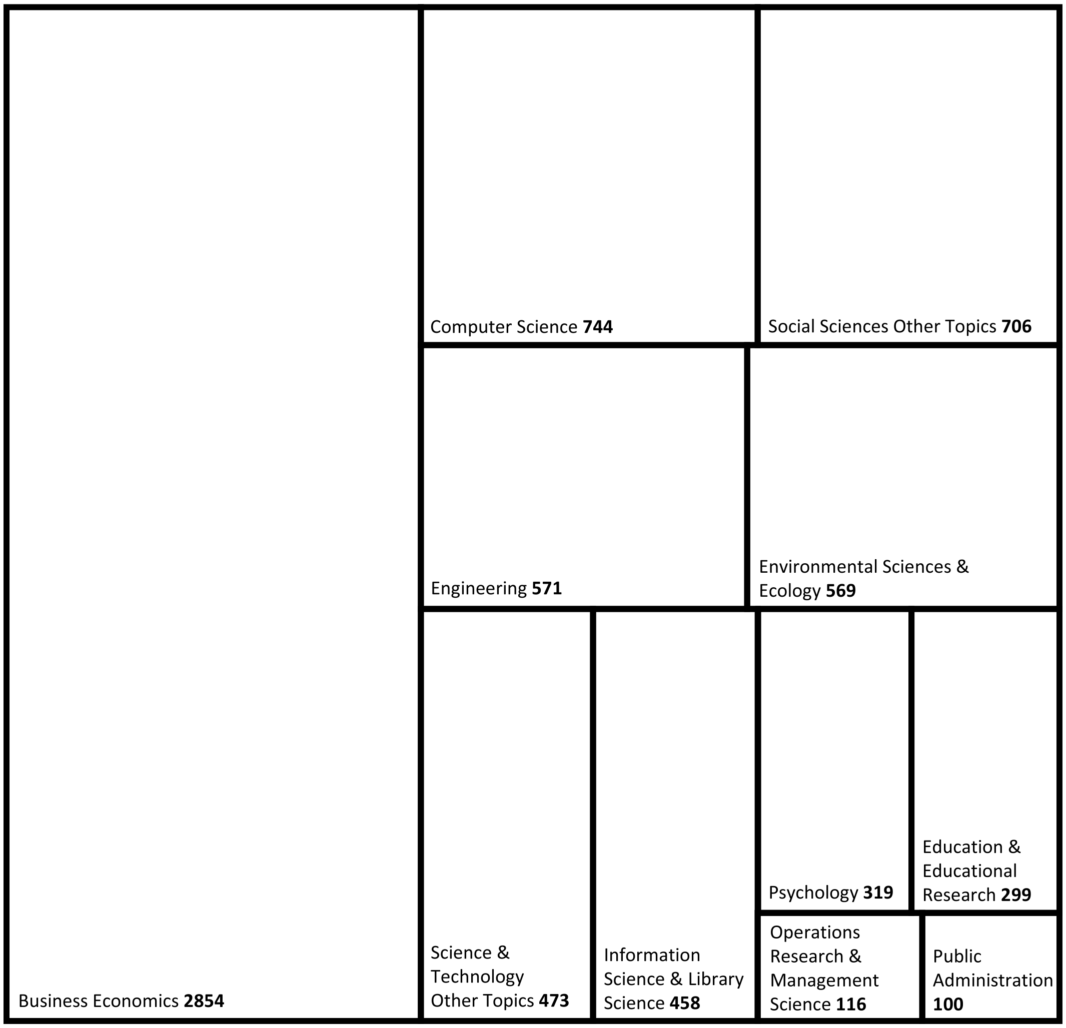

Figure 1. Tree map based on the web of science citation numbers of Hair et al. (2011) article in different research disciplines. Note: Figure prepared on January 21, 2022; numbers in bold are citation counts.

featured quite prominently in marketing research (Sarstedt et al., 2022a) and other fields of scientific inquiry (Hair et al., 2022, Chapter 1). Your paper therefore tackles a relevant topic and has the potential to offer guidance for future use of the method. Having said this, while many of your initial descriptions accurately reflect the latest thinking in the field, some of the article contents rely on prior understanding of the method which has been challenged and revised. For example, several of the metrics for measurement and structural model assessment have been replaced with more effective criteria. In other instances, your descriptions should be more precise, such as when discussing structural model assessment in terms of its "predictive relevance" (page 147). In the following, we will focus on these critical aspects of your paper and highlight recent developments in the field. We structure our discussions around the following three topics: (1) PLS-SEM's characteristics and distinguishing features, (2) model evaluation, and (3) bootstrapping.

PLS-SEM's characteristics and distinguishing features

In the first part of your paper, you describe the principles of SEM and discuss the key differentiating factors between PLS-SEM and CB-SEM. While most of your descriptions have merit, others lack clarity or are outdated. Let us be specific.

Your discussion of the PLS-SEM bias implies that CB-SEM produces the correct results by which PLS-SEM should be benchmarked. This notion is problematic, however, since it rests on assumptions about the nature of the constructs, which are – by definition –

unknown. Why should one consider the common factor model as the gold standard to estimate constructs, as implied by the CB-SEM method? In fact, researchers in various fields of science have shown increasing appreciation that common factors may not always be the right approach to measure concepts (e.g. Cho et al., 2021b; Rhemtulla et al., 2020). Recent research also indicates common factor models can be subject to considerable degrees of metrological uncertainty with serious consequences regarding their validity (Rigdon & Sarstedt, in press; Rigdon et al., 2020). Apart from the above, researchers should rather ask what the consequences are of misapplying each method on the wrong model. This is also what you indicate in your paper when stating that using PLS-SEM on common factor model data produces biases that are of "minor relevance for practical applications" (page 143). In fact, researchers have repeatedly illustrated that bias resulting from using PLS-SEM to estimate common factor models is generally trivial. One example is Sharma et al.'s (in press-b) study, showing recent claims that seminal information systems models lack validity because they were estimated using PLS-SEM (Evermann & Rönkkö, in press), are grossly exaggerated and not supported by any empirical evidence. More importantly, research has shown the biases produced by CB-SEM when using the method on composite model data is far more pronounced than the "PLS-SEM bias" (Cho et al., 2021a; Sarstedt et al., 2016). PLS-SEM seems to be the safer choice, therefore, when the underlying model type is unknown. Your paper should reflect these aspects.

Related to the previous point, you should more clearly distinguish reflective and formative measurement from common factor and composite models. Several researchers equate these concepts, thereby causing much confusion among users of PLS-SEM (Rigdon et al., 2014). Reflective and formative measurements describe the relationships between constructs and their indicator measures from a measurement theoretic perspective. The choice of measurement specification is a conceptual and theoretical one – as extensively documented in the literature (e.g. Bollen & Bauldry, 2011; Jarvis et al., 2003) – and is unrelated to how the constructs are treated in a statistical model and their scores being estimated (i.e. using Mode A or B). Common factor and composite models on the other hand describe different approaches for representing constructs in a model. Equating reflective measurement with common factors is not appropriate as this measurement type may also be adequately represented by composites (see Sarstedt et al., 2016). This is also why calls for universal application of methods which mimic maximum-likelihood CB-SEM results when estimating reflective models – such as consistent PLS-SEM (Dijkstra, 2014; Dijkstra & Henseler, 2015) and efficient PLS-SEM estimation (Bentler & Huang, 2014; Huang, 2013) – should be considered with great caution.

You also note that CB-SEM is the appropriate method for theory testing while PLS-SEM is better suited for theory development and prediction (page 140). This notion is somewhat odd, considering PLS-SEM is routinely used to test models whose relationships reflect hypotheses that researchers developed based on theory and logic. Of course, PLS-SEM is appropriate for testing theories. However, different from CB-SEM which assumes an explanatory perspective, PLS-SEM is causal-predictive (Wold, 1982) in that the method emphasizes (in-sample) prediction in estimating statistical models, whose theoretical structures are designed to provide causal explanations (Chin et al., 2020). Establishing the predictive power of structural equation models is important as this step safeguards their practical utility (Sarstedt & Danks, 2021). At the same time, prediction without explanation – as executed by, for example, machine learning procedures (Hair & Sarstedt, 2021a) – casts doubt on the generalizability of the corresponding models and their boundary conditions (Sharma et al., in press-b).

SEM algorithm and its objectives is imprecise. You only refer to its objective being to maximize "the explained variance of the dependent latent constructs" (page 139); that is, the "R^2 values" (page 148). However, PLS-SEM not only maximizes the explained variance in the structural model but also in the measurement models (i.e. the indicator's communality). While this characteristic has been extensively documented in the literature (e.g. Chin, 1998; Lohmöller, 1989; Tenenhaus et al., 2005; Wold, 1982), critics such as Rönkkö et al. (in press) interpret your somewhat imprecise descriptions in an attempt to make a case against PLS-SEM on quite dubious grounds. Your original description of the PLS-SEM algorithm could also be extended to include more recent advances like Becker and Ismail`s (2016) weighted PLS (WPLS). This method enables researchers to incorporate sampling weights into the standard PLS-SEM algorithm that assign a different relevance to observations, depending on whether the observations are under- or over-represented in the sample compared to the population. Cheah et al. (2021) provide additional details on WPLS and its applications.

When discussing differences between the two SEM methods, you mention "there is no adequate measure of goodness of fit" in PLS-SEM (page 143), but offer no further details in this regard. In fact, recent research has witnessed substantial debates regarding whether or not model fit metrics – as proposed in a CB-SEM context –

are applicable to PLS-SEM. For example, Hubona et al. (2021) stress the importance of the model fit concept in PLS-SEM and call for the routine use of SRMR and a statistical test of distance-based metrics such as d_U and d_G, which have been conceptually discussed (Schuberth & Henseler, in press) and empirically analyzed (Schuberth et al., 2018). Other researchers contend these metrics are largely unsuitable for model fit testing in research settings commonly encountered in applied research (Hair et al., 2022, Chapters 4, 6). Most scholars seem to vote with their own feet as very few researchers apply these model fit metrics (Sarstedt et al., 2022a). Contributing to the widespread reluctance of adopting model fit metrics to PLS-SEM might be that the various criteria advocated in extant research often produce conflicting results. For example, while the SRMR might indicate a well-fitting model, one or both distance-based metrics indicate a lack of fit, leaving users in doubt. Also, a well-fitting model (e.g., in terms of SRMR) does not necessarily offer adequate predictive power (Sarstedt & Danks, 2021). Although the application of model fit criteria in the context of composite models requires additional research (Cho et al., 2020, in press), a foundational article like yours on PLS-SEM should include these different perspectives.

Another distinguishing feature you mention between PLS-SEM and CB-SEM is the availability and application of construct scores. Specifically, your notion that CB-SEM does not estimate "stable" scores (page 143) needs further clarification. What exactly do you mean by "stable"? Construct (or factor) score indeterminacy in CB-SEM implies that a common factor cannot be faithfully reproduced as one unique best function of the observed variables in its measurement model (Rigdon et al., 2019). You should explain this notion and also discuss its implications for methods use. For example, Schönemann and Haagen (1987) empirically show that construct scores can always be chosen such that they predict any criterion perfectly, regardless how it correlates with the predicting constructs' indicators.

You also note that PLS-SEM-based construct scores are "more precise." But what makes these scores more precise considering the entities under consideration are unobservable? Analogous to George Box's famous quote "All models are wrong, but some are useful," a certain set of construct scores should be considered as *useful* for representing conceptual variables in the PLS path model rather than "more precise." To address this question, you should more carefully distinguish between conceptual variables, constructs and observed variables (Hair et al., 2017a; Rigdon, 2012). Doing so would help to better pinpoint the validity gaps that may emerge in PLS-SEM and CB-SEM analyses (Hair & Sarstedt, 2019).

You also highlight some rules of thumb for selecting CB-SEM or PLS-SEM in a separate table. Apart from the points mentioned above, some elements of this table should be updated and clarified.

- You recommend the ten times rule for determining the minimum sample size, which has been shown to be ineffective (Kock & Hadaya, 2018; Marcoulides & Saunders, 2006; Rigdon, 2016). While it may offer rough guidance, solely relying on this rule is inadequate and perpetuates research practices that have been criticized for good reason (Petter, 2018). Instead, you should call for the use of power analyses, which have been documented in tutorials (Aguirre-Urreta & Rönkkö, 2015; Hair et al., 2022, Chapter 5). Alternatively, researchers may draw on Kock and Hadaya's (2018) inverse square root method, which is relatively easy to apply and offers a more precise picture of minimum sample size requirements than the ten times rule does.
- Maximum likelihood-based CB-SEM estimation is fairly robust against violations of normality, as long as the sample size is large relative to model complexity (Reinartz et al., 2009). Hence, nonnormal data is not a strong argument for preferring PLS-SEM over CB-SEM unless larger samples are not available for your research context.
- You note that researchers should apply CB-SEM if measurement invariance needs to be established. However, Henseler et al. (2016) proposed the measurement invariance of composite models (MICOM) procedure, which facilitates such analyses as a requirement before conducting the bootstrap-based multigroup analysis (Sarstedt, Henseler, et al., 2011) and permutation based multigroup assessment (Chin & Dibbern, 2010) in PLS-SEM (Matthews, 2017).
- Building on its causal-predictive nature, another meaningful reason for using PLS-SEM is the method's efficacy for running predictive model comparisons. For example, Sharma et al. (2021) have identified information-theoretic model selection criteria that allow selecting well-fitting models that also excel in terms of prediction. In addition, Liengaard et al.'s (2021) cross-validated predictive ability test (CVPAT) and its extension (Sharma et al., in press-a) facilitate model comparisons from a prediction-only perspective.

A final and more general concern is that your guidelines on "choosing PLS-SEM versus CB-SEM" (page 143) likely lead to black-and-white thinking, which critics of PLS-SEM readily echoe in recent years (e.g. Rönkkö et al., 2016). We understand that applied users need concrete guidance as to whether PLS-SEM or CB-SEM is the preferred method for their research situation. However, as called for in the past decade by Rigdon (2012), the PLS-SEM method has clearly been emancipated from its covariance-based counterpart. Hence, there is no need to treat the method as an "attractive alternative" (page 140) to the alleged standard CB-SEM approach – see, Rigdon et al. (2017). This particularly holds since researchers have long argued the two methods cannot be practically compared, because of inherent differences in their functioning principles (Lohmöller, 1989; Marcoulides et al., 2012; Rigdon et al., 2017; Schneeweiß, 1991). With their different statistical objectives and methodological requirements, the PLS-SEM and CB-SEM methods are complementary rather than competitive for estimating latent variable models (Jöreskog & Wold, 1982). Researchers should therefore emphasize PLS-SEM's causal-predictive nature and its ability to obtain meaningful solutions in complex models with many indicators and relatively small sample sizes (Hair et al., 2019c).

Model evaluation

Reflective measurement model evaluation

Your descriptions of reflective measurement model evaluation have merit but do not reflect the state of the art in the field. While you correctly note that the tau equivalence assumption (i.e. equal population indicator loadings) of Cronbach's alpha is typically violated, the statistic still offers value in a PLS-SEM context as it can be regarded as a conservative construct measure's internal consistency reliability, thereby representing a lower reliability boundary. On the contrary, composite reliability ρ_C that you favor, is quite liberal, thereby representing the upper reliability boundary. As an alternative, Dijkstra (2014) proposed the approximate exact (or consistent) reliability coefficient ρ_A (see also Dijkstra and Henseler, 2015), which explicitly takes the indicator weights estimates into account. The reliability coefficient ρ_A usually lies between the more conservative Cronbach's α and the more liberal composite reliability ρ_c, and is therefore considered a good compromise between these other two measures (Sarstedt et al., 2021).

More importantly, your call to assess the measures' discriminant validity using the Fornell Larcker criterion (Fornell & Larcker, 1981) and the indicator cross loadings is outdated. Recent research challenges the efficacy of the Fornell-Larcker criterion on both empirical and conceptual grounds (Franke & Sarstedt, 2019; Hair et al., 2020a; Radomir & Moisescu, 2020). For example, Henseler et al. (2015) show the criterion does not reliably detect a lack of discriminant validity when indicator loadings are relatively homogeneous (e.g. the loadings vary between 0.60 and 0.80) – as is typically the case in applications of PLS-SEM. The authors also show that cross loadings (i.e. high indicator correlations with an opposing construct) perform even worse – they do not identify substantial violations of discriminant validity. Hence, your notion that cross loadings are "a bit more liberal" (p. 146) has been confirmed by recent research as a substantial understatement since this criterion almost never indicates a lack of discriminant validity.

Instead, researchers should apply Henseler et al.'s (2015) heterotrait-monotrait ratio (HTMT), which contrasts the correlations of indicators measuring different constructs with the correlations of indicators measuring the same construct. An HTMT value of close to 1 (or higher) indicates a lack of discriminant validity. The HTMT criterion has been validated in subsequent simulation studies (Franke & Sarstedt, 2019; Voorhees et al., 2016) and has become the standard criterion for discriminant validity assessment in PLS-SEM. Recent guidelines articles also emphasize the importance of testing whether the HTMT values are significantly lower than a prespecified threshold such as 0.85 or 0.90 (Hair et al., 2022, Chapter 4). Specifically, researchers should assume a lower (i.e. more conservative) threshold such as 0.85 when comparing conceptually distinct constructs. When the constructs to be compared are conceptually similar (e.g. cognitive and emotional satisfaction), a threshold of 0.90 or even closer to 1 should be used.

Finally, research has shown the singular assessment of the average variance extracted as an indication of convergent validity for reflective measurement models can fail to identify model misspecifications (Rönkkö & Evermann, 2013; Rönkkö et al., in press). You should, therefore, offer more details on how to apply the model evaluation metrics in practice. For example, your comments should (1) emphasize that researchers should consider several criteria for model assessment jointly in order to disclose potential misspecifications, as described in various textbooks and guideline articles on PLS-SEM use (e.g. Hair et al., 2020b, 2022; Ramayah et al., 2016), and (2) stress that content validity considerations play a pivotal role in model assessment (Hair et al., 2021).

Formative measurement model evaluation

Many of your descriptions of formative measurement model evaluation reflect current thinking. For example, your differentiation of the relative and absolute contributions of the indicators in forming a construct is a valuable metric. Other assessment metrics are incomplete, however, and somewhat ambiguous.

An important step in formative measurement model assessment is establishing the measure's convergent validity on the grounds of a redundancy analysis (Chin, 1998). This approach involves correlating the formative construct with an alternative reflective or single-item measure of the same concept (see Cheah et al., 2018 for a discussion). A correlation of 0.70 or higher would indicate a sufficient level of convergent validity. Note that to execute a redundancy analysis, additional global indicators must be included in survey research. With secondary data, alternative measures will have to be identified from related secondary sources.

In addition, while your notion that heterogeneous response patterns may cause nonsignificant indicator weights has merit, finite mixture PLS (FIMIX-PLS; Hahn et al., 2002) is not an effective latent class procedure to resolve this issue – for at least two reasons. First, FIMIX-PLS is not capable of handling heterogeneity in the measurement models as the method uses construct scores produced by the standard PLS-SEM analysis as input for a series of finite mixture regressions to capture heterogeneity in the structural model only. Second, while FIMIX-PLS is capable of identifying heterogeneous response patterns (Sarstedt, Becker, et al., 2011), the method has been largely incapable of reproducing the correct data partition (Ringle et al., 2014). Researchers have therefore called for the joint application of FIMIX-PLS and Becker et al.'s (2013b) PLS prediction-oriented segmentation method (PLS-POS), or Schlittgen et al.'s (2016) iterative reweighted regressions segmentation method for PLS (PLS-IRRS), which reliably handle heterogeneity in both measurement and structural models (Sarstedt et al., 2022b).

Apart from the above, latent class analyses should not be confined to situations where formative measurement model assessment produces a series of nonsignificant indicator weights. Instead, they should be an integral part of any PLS-SEM analysis as a type of robustness check (Sarstedt et al., 2020b) – as suggested on page 147 of your paper.

Finally, you correctly point to the need to check for collinearity issues in formative measurement models. However, your suggested threshold value for the variance inflation factor (VIF) of 5 is too liberal. More recent research emphasizes that collinearity-induced problems such as inflated standard errors or sign changes in the indicator weights can occur at much lower VIF levels (Hair et al., 2022, Chapter 5). Researchers should therefore assume a VIF threshold of 3 as indicative of potential collinearity issues. Having said this, the impact of collinearity on PLS-SEM results and identifying flexible cutoff values for the VIF that consider a variety of data and model constellations still needs to be explored and confirmed with additional methodological research.

Structural model evaluation

While explanatory power refers to "the strength of association indicated by a statistical model" (Shmueli & Koppius, 2011, p. 561), predictive power refers to the model's ability to predict outcome values based on previously unobserved data, typically drawn from hold out samples. Sarstedt and Danks (2021) show that a model with a certain level of explanatory power can produce vastly different levels of predictive power and vice versa (see also, Shmueli, 2010). Hence, these two dimensions of structural model evaluation need to be clearly differentiated. This is not the case, however, in your paper.

For example, your description of the R^2 does not identify this metric as solely a measure of explanatory power (in-sample prediction) that uses the entire dataset to explain the endogenous constructs' case values (Shmueli & Koppius, 2011). Apart from this, your descriptions of what constitutes substantial, moderate, and weak R^2 values is oversimplified for several reasons. First, the R^2 metric is a function of model complexity – as the number of antecedent constructs in your model increases, so does the R^2. Second, R^2 values depend on the phenomenon being explained. While high R^2 values of 0.90 might be plausible when explaining physical processes, this is certainly not the case when dealing with human attitudes, perceptions, and intentions where such values often indicate model overfit (Hair et al., 2019b; Hair & Sarstedt, 2021b).

Equally important, your description of the blindfolding-based Q^2 as a metric of "predictive relevance" (page 147) is misleading. Apart from the fact you do not clearly define the concept of "predictive relevance," the blindfolding-based Q^2 is in fact not a prediction metric. The blindfolding procedure systematically imputes singular points in the data matrix with the mean value, which are then being predicted using the PLS-SEM estimates as input. This approach leaves the general sample structure intact, making the blindfolding-based Q^2

a questionable hodgepodge of an explanation and prediction metric (Shmueli et al., 2016). Instead, researchers should apply Shmueli et al.'s (2016) $PLS_{predict}$ procedure, which implements *k*-fold cross-validation in a PLS-SEM context.

The $PLS_{predict}$ procedure divides the sample into a prespecified number of *k* subsets of equal size. $PLS_{predict}$ then combines *k*-1 subsets to estimate the parameters, which are then being used to predict the case values of the left-out subset (i.e. the holdout sample). The process is repeated *k* times such that each subset serves as the holdout sample once. Researchers then assess the PLS path model's predictive power by comparing its prediction error – typically quantified by means of the root mean squared error – with that of a naïve mean value prediction benchmark ($Q^2_{predict}$) and that of a more demanding linear model (LM) benchmark (Shmueli et al., 2019). Note that this process of predictive power assessment can also be generalized to more complex model set-ups involving mediation effects (Danks, 2021). However, $PLS_{predict}$ does not offer a statistical test to determine whether a proposed model offers a significantly better out-of-sample predictive accuracy than a benchmark. Addressing this concern, Sharma et al. (in press-a) recently introduced a test for assessing the stand-alone predictive power of a model.

To summarize, we believe that establishing a PLS path model's predictive power requires much more attention in your description of structural model assessment. PLS-SEM's prediction orientation is a key differentiating factor from CB-SEM, which is solely concerned with explanation (Hair & Sarstedt, 2021b; Legate et al., 2022; Manley et al., 2021; Richter, Cepeda Carrión, et al., 2016; Sarstedt & Danks, 2021).

Finally, your descriptions implicitly suggest researchers should examine a single model only. While this is still the common practice in marketing and other fields of business research (Sharma et al., 2019), you should encourage researchers to hypothesize different theoretically justified model configurations and compare them using model selection criteria (e.g. see Nuzzo, 2015). Sharma et al. (2019, 2021) have compared the efficacy of different criteria in a PLS-SEM context and found the Bayesian information criterion (Schwarz, 1978) and the Geweke-Meese criterion (Geweke & Meese, 1981) perform well in identifying a model that fits the data well. Transforming the raw criterion values into Akaike weights offers evidence for each model's relative likelihood (Danks et al., 2020). Finally, Liengaard et al.'s (2021) CVPAT enables testing whether an alternative model has significantly better predictive capabilities than an established model. These prediction-oriented model comparison options must be considered by you to adequately support researchers in their decision-making.

Bootstrapping

Your discussion of bootstrapping is very useful, especially since many applied researchers are unaware of its relevance in a nonparametric methods context. While your description is certainly meant as a generic introduction into the concept, more details regarding its applicability in boundary conditions may be needed in light of recent developments in this area. Specifically, it would be worthwhile to explicitly note that bootstrapping reliably identifies null effects (Henseler et al., 2014), despite contrary claims by critics of the PLS-SEM method (e.g. Rönkkö et al., in press).

We also have two further comments regarding the application of bootstrapping. First, rather than advocating the use of bootstrapping solely for deriving *t*-values, you should encourage the computation of bootstrap-based confidence intervals. Aguirre-Urreta and Rönkkö (2018) have compared the efficacy of different methods for constructing confidence intervals, showing the percentile method generally performs best. But for highly asymmetric distributed parameters, the bias-corrected and accelerated bootstrap confidence intervals may be used. Second, your paper should reflect more recent recommendations by Streukens and Leroi-Werelds (2016), indicating that bootstrapping should be executed with 10,000 subsamples.

Looking ahead

Like all statistical methods, PLS-SEM relies on continual fine-tuning and gradual extensions to ensure its applicability in various research contexts (Hair et al., 2020a, 2021; Legate et al., 2022; Manley et al., 2021; Sarstedt et al., 2014; Sharma et al., in press-b). Such progress requires continuous critical reflection on whether applied researchers follow best practices (Kreamer et al., 2021), but also whether seminal papers, published years ago, accurately reflect the latest thinking in the field. Otherwise, misapplications may be perpetuated as researchers rely on outdated concepts and descriptions, which, at the time of Hair et al.'s (2011) writing, were adequate, but not when applying today's knowledge and standards. Critics have referred to outdated descriptions to make a case against PLS-SEM, often readily ignoring more recent updates to the method. For example, Rönkkö et al. (in press) draw on the Fornell-Larcker criterion to show that PLS-SEM is allegedly incapable of detecting model misspecifications. But they ignore the fact that (1) this criterion has long been debunked as ineffective, and (2) researchers have frequently called for the use of the HTMT statistic in discriminant validity assessment (e.g. Hair et al., 2019b; Henseler et al., 2015; Sarstedt et al., 2021).

Critical accounts of prior research, regardless of whether targeted at a method, theory, or concept, need to be grounded in the most recent research – any other practice runs contrary to the fundamental quest of science. Review studies play an important role in documenting the current state of a method's application in specific disciplines in an effort to uncover areas for improvement. For PLS-SEM, such review studies have been performed in, for example, accounting (Lee et al., 2011; Nitzl, 2016), (international) management (Hair et al., 2012a; Richter, Sinkovics, et al., 2016), knowledge management (Cepeda Carrión et al., 2019), and marketing (Hair et al., 2012b; Sarstedt et al., 2022a).

In Table 1, we summarize the aspects covered in Hair et al. (2011) that require an update, clarification, or extension based on more recent methodological research.

Scientific progress is a million little steps and as our knowledge regarding PLS-SEM's strengths and limitations accumulates, users of the method also need to consider the latest guidelines for correctly applying the method. Progress is not linear, however, since researchers develop different understandings of concepts and methods and sometimes advocate alternative approaches. One recent example in the field is the controversy surrounding the confirmatory composite analysis (CCA) approach, which describes a set of procedures used to validate composite models estimated by PLS-SEM. Specifically, there is substantial disagreement as to which analysis steps define a CCA (e.g. Hubona et al., 2021) and how that process was developed (Hair et al., 2020b). While Schuberth et al.'s (2018) approach strongly emphasizes model fit, this is not the

Table 1. Latest thinking and clarifications regarding PLS-SEM.

Prior thinking	Latest thinking/Clarification/Correction	References
PLS-SEM's characteristics and distinguishing features		
Common factor models are the standard against which PLS-SEM estimates should be benchmarked	Reducing SEM to common factor models is highly restrictive since most concepts can also be represented by composite models. Researchers should closely assess the biases that occur when using a method on the wrong model type.	Cho et al. (2021a); Hair et al. (2017b); Rigdon et al. (2017)
Reflective measurement equals common factor models, formative measurement equals composite models	The measurement-theoretic perspective (i.e. specifying models as reflective or formative) needs to be separated from how measures are treated in a statistical model. Reflective measures can also be estimated using composite models.	Sarstedt et al. (2016)
SEM should only be concerned with explanation	Establishing a model's predictive power is a critical test of the relevance of explanation and a fundamental requirement for deriving actionable recommendations from the results.	Hair and Sarstedt (2021b); Sarstedt and Danks (2021)
LS-SEM is particularly suitable for exploratory research, while CB-SEM should be used for theory testing	Both methods allow for testing theories. In doing so, PLS-SEM follows a causal-predictive paradigm, while CB-SEM is largely unsuitable for predictive purposes.	Becker et al. (2013a); Legate et al. (2022); Sharma et al. (in press-b)
Use the ten times rule for determining the minimum sample size for PLS-SEM analyses	The ten times rule has been debunked as ineffective. Researchers should run power analyses or apply Kock and Hadaya's (2018) inverse square root method.	Aguirre-Urreta and Rönkkö (2015); Kock and Hadaya (2018)
Researchers should use PLS-SEM to handle nonnormal data	Maximum likelihood-based CB-SEM is fairly robust against violations of normality with large sample sizes. Preferring PLS-SEM over CB-SEM solely on the grounds of nonnormal data is not a valid argument.	Hair et al. (2017a); Reinartz et al. (2009)
Measurement invariance assessment can only be executed with CB-SEM	The MICOM procedure allows assessing measurement invariance testing in PLS-SEM.	Henseler et al. (2016)
PLS-SEM is an attractive alternative to the standard CB-SEM method	PLS is a full-fledged SEM estimator, which offers much flexibility and produces lower biases in the estimation of latent variable models whose nature (i.e. common factor or composite model) is, by definition, unknown.	Petter and Hadavi (2021); Rigdon (2012); Rigdon et al. (2017)
Model evaluation		
Reflective measurement model evaluation		
Internal consistency reliability should rely on Cronbach's alpha and composite reliability ρ_C	Cronbach's alpha constitutes the lower boundary, while ρ_C the upper boundary of internal consistency reliability. Researchers should also consider the ρ_A as a good compromise between these two metrics.	Dijkstra and Henseler (2015); Sarstedt et al. (2021)
Discriminant validity assessment should rely on the Fornell-Larcker criterion and an assessment of cross loadings	The Fornell-Larcker criterion and cross loadings are ineffective for discriminant validity assessment. The HTMT criterion should be used instead, preferably in an inferential test.	Franke and Sarstedt (2019); Henseler et al. (2015)

(*Continued*)

Table 1. (Continued).

Prior thinking	Latest thinking/Clarification/Correction	References
Formative measurement model evaluation		
Model evaluation focuses on the indicators' relative and absolute contributions as well as indicator collinearity	Model evaluation also needs to consider the formative measures' convergent validity by applying the redundancy analysis.	J.-H. Cheah et al. (2018); Chin (1998, 2010)
Use FIMIX-PLS to assess whether measurement model estimates are adversely affected by unobserved heterogeneity	FIMIX-PLS only captures heterogeneity in the structural model estimates. Researchers should use more advanced latent class techniques such as PLS-IRRS and PLS-POS.	Becker et al. (2013b); Sarstedt et al. (2022b); Schlittgen et al. (2016)
VIF values above 5 are indicative of collinearity	Collinearity issues can occur at lower VIF levels between 3 and 5. More research is needed, however, to establish firm guidelines.	Mason and Perreault (1991)
Structural model evaluation		
The R^2 indicates a model's predictive power	The R^2 is an indicator of explanatory and not predictive power. Its main value is identifying model overfit.	Hair et al. (2019b); Hair et al. (2019c); Shmueli and Koppius (2011)
The blindfolding-based Q^2 metric should be used to assess a model's "predictive relevance"	The blindfolding-based Q^2 is not a reliable indicator of explanatory or predictive power. Explanatory power assessment should rely on the R^2, while $PLS_{predict}$ or CVPAT should be used for predictive power assessment.	Sharma et al. (in press-a); Shmueli et al. (2016); Shmueli et al. (2019)
Researchers should analyze a single model only	Researcher should hypothesize and estimate different theoretically justifiable model configurations and compare them using model selection criteria and/or a statistical test.	Liengaard et al. (2021); Sarstedt and Danks (2021); Sharma et al. (in press-a); Sharma et al. (2019)
Bootstrapping		
Use bootstrapping to derive *t*-values	Researchers should use bootstrap confidence intervals for inference testing.	Aguirre-Urreta and Rönkkö (2018)
Bootstrapping should be run with 5,000 subsamples	Researchers should use 10,000 subsamples in bootstrapping.	Streukens and Leroi-Werelds (2016)

case with Hair et al.'s (2019a; 2020b) CCA procedure that rather focuses on the assessment of the model's predictive power. Users of PLS-SEM should carefully consider opposing positions and weigh their arguments in order to make an informed decision as to which way to proceed (Hair, 2021). Researchers acting out these conflicts should remind themselves that any position, including their own, comes with uncertainties – "my way or the highway" thinking or efforts to "set the record straight" have no place in serious science.

Our discussion of Hair et al. (2011) did not consider the article's subtitle "a silver bullet," which likely contributed to its success (Sarstedt, 2019). Researchers working in the field will know that the subtitle refers directly to Marcoulides and Saunders' (2006) editorial titled "PLS: A Silver Bullet?" in which the authors reflect critically on the common belief that PLS-SEM can always be used when the sample size is small. Similarly, Sosik et al. (2009, p. 5) ask whether PLS-SEM should be considered "silver bullet or voodoo statistics." Hair et al. (2011) not only tie in with these papers but also make clear that PLS-SEM does not offer an inerrant means for estimating latent variable models. Indeed, Hair et al. (2011) emphasize the opposite when stressing that "we do not argue against the statistically more demanding CB-SEM method. CB-SEM and PLS-SEM are complementary in that the nonparametric and variance-based PLS-SEM approach's advantages are the parametric and covariance-based SEM approach's disadvantages and vice versa" (Hair et al., 2011, p. 149). Yet, the notion that Hair et al. (2011) unequivocally refer to PLS-SEM as a "silver bullet" features prominent in many recent critiques of the method (Evermann & Rönkkö, in press; Rönkkö et al., in press) where researchers use this term as a tagline to portray us as blindly advocating a supposedly inferior method. Similarly, but in a very balanced account of PLS-SEM, Russo and Stol (in press) note that describing the method as a silver bullet is "perhaps, too optimistic." We concur with this notion and their conclusion that "in every statistical analysis, every methodological choice comes with affordances, but also at a cost."

We firmly believe composite-based SEM in general and particularly PLS-SEM offers many advantages over factor-based methods such as CB-SEM. These advantages include, but are not limited to the ability to estimate complex models, the inherent emphasis on a rigorous prediction perspective, and achieving high levels of statistical power. At the same time, we agree with Petter and Hadavi (2021, p. 10) that "with the great power of PLS also comes great responsibility." Living up to this responsibility not only requires offering solid arguments for using the method and following the most recent guidelines for its use (e.g. Hair et al., 2022), but also applying state-of-the-art methodological extensions of the basic PLS-SEM method. For example, recent research has made considerable improvement in the treatment of observed heterogeneity (Klesel et al., 2019) and latent class analysis (Sarstedt et al., 2022b). Another stream of research introduces the necessary condition analysis to PLS-SEM, which implies that an outcome or a certain level thereof can only be achieved if the necessary cause is in place, or at a certain level (Richter et al., 2020). Progress has also been made in the specification, estimation, and validation of higher order models (Sarstedt et al., 2019a, as well as predictive model assessment (Sharma et al., in press-a), mediation (Cheah et al., 2021; Memon et al., 2018; Sarstedt et al., 2020a), moderation (Becker et al., 208; Memon et al., 2019), nonlinear effects (Basco et al., 2021), and missing value treatment (Wang et al., 2022).

With these extensions PLS-SEM has become a full-fledged composite-based estimator for latent variable models whose scope greatly extends beyond what Hair et al. (2011) documented a decade ago. In the future, important methodological advances will ensure the dynamic further development of the PLS-SEM method and thus its increasing relevance for science and practice. Examples of such advances might include the estimation of multilevel models (Hwang et al., 2007), the handling of archival, choice, longitudinal and panel data (Jung et al., 2012; Roemer, 2016), the assessment of endogeneity using Gaussian copulas (Becker et al., 2022; Hult et al., 2018; Park & Gupta, 2012), the philosophical differences between common factor and composite models and their empirical identification (e.g. by using a confirmatory tetrad analysis; Bollen & Ting, 2000; Gudergan et al., 2008; Ryoo & Hwang, 2017), and the evaluation of common method variance (e.g. Chin et al., 2012, 2013).

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